INTERNATIONAL MONETARY FUND
AND INTERNATIONAL DEVELOPMENT ASSOCIATION

BURUNDI

Joint IMF/World Bank Debt Sustainability Analysis 2011

Prepared by the Staffs of the International Monetary Fund
and the International Development Association

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Given its narrow export base and low capacity of government institutions, Burundi's risk of external debt distress remain high, unchanged from the 2010 debt sustainability assessment. The present value (PV) of debt-to-exports ratio exceeds the policy dependent threshold in the medium-term, while alternative scenarios and stress tests suggest that two debt indicators will breach indicative thresholds. The public DSA suggests that while Burundi is in a position to service its public and publicly guaranteed debt, a lower growth would lead to a considerable deterioration of debt indicators. Burundi should continue to implement prudent borrowing policies and accelerate structural reforms for private sector led growth, particularly in infrastructure and export diversification, to mitigate risks to debt sustainability.[1]

I. BACKGROUND

1. **The last Low Income Country Debt Sustainability Analysis (LIC DSA) conducted in 2010 concluded that Burundi would continue to face a high risk of debt distress.**[2] The PV of debt-to-exports ratio exceeded the country-specific policy-dependent threshold by a wide margin throughout the projection period, reflecting the country's narrow export base. In addition, under alternative scenarios and related stress tests, all but one debt indicators exceeded the country-specific debt burden thresholds.

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[1] The DSA has been produced jointly by IMF and World Bank staffs.

[2] See IMF Country Report No.10/313, Sup. 2; and IDA/R2010-0346/1, Annex 2.

2. **Burundi is a weak policy performer for the purpose of determining the debt burden thresholds under the Debt Sustainability Framework (DSF).** Burundi's rating on the World Bank's Country Policy and Institutional Assessment (CPIA) averaged 3.07 (on a scale of 1 to 6) during 2008–10, making it a weak policy performer.[3]

3. **Burundi's stock of external public and publicly guaranteed (PPG) debt has declined significantly since 2008.** This reflects debt relief under the enhanced Highly Indebted Poor Countries (HIPC) Initiative and the Multilateral Debt Relief Initiative (MDRI). Nominal external PPG debt amounted to US$455 million or 22 percent of GDP in 2010 while the PV of debt-to-exports ratio stood at 150.8 percent.[4] About 90 percent of Burundi's outstanding nominal external PPG debt was owed to multilateral creditors, with bilateral creditors accounting for the remainder. Efforts to secure HIPC Initiative debt relief from all creditors are continuing.[5]

Text Table 1. Burundi: Public External Debt, 2008-10
(Millions of US dollars)

	2008			2009			2010		
	Nominal Stock	Percent of GDP	Percent of Total	Nominal Stock	Percent of GDP	Percent of Total	Nominal Stock	Percent of GDP	Percent of Total
Total Debt	1,518	98	100.0	376	21	100.0	455	22	100.0
Multilateral	1,313	84	86.5	338	19	89.9	411	20	90.2
Bilateral	205	13	13.5	38	2	10.1	45	2	9.8
Paris club	151	10	10.0	1	0	0.4	1	0	0.3
Non-Paris club	53	3	3.5	36	2	9.7	43	2	9.5
Commercial	0	0	0.0	0	0	0.0	0	0	0.0

Sources: Burundian authorities and Bank-Fund staffs estimates.

[3] Burundi showed a slight improvement in its CPIA score from the 3.05 averaged during 2007–09. A score below 3.25 corresponds to a poor policy performance, according to the LIC Debt Sustainability Framework (DSF).

[4] General government, on a gross basis. As noted in the Technical Memorandum of Understanding, debt contracted by state-owned enterprises (SOEs) with a government guarantee is included in debt limits and therefore in the DSA.

[5] Among multilateral creditors, agreements have been signed with BADEA, the EU, and the OFID Fund, while negotiations are ongoing with IFAD. Among non-Paris Club creditors, agreements have been signed with Kuwait, Libya, Saudi Arabia and the UAE. The People's Republic of China cancelled all of its claims during the interim period. The terms of some of the signed agreements were less favorable than those envisaged under the HIPC Initiative.

4. **Domestic debt comprises a significant component of total public debt.** About 40 percent of Burundi's total public debt was held by domestic creditors in 2010, representing about 15 percent of GDP, up from 11.5 percent in 2008. The main reason for the increasing importance of domestic debt was the cancellation of external public debt as a result of HIPC Initiative and MDRI debt relief. Most domestic public debt is owed to the central bank and has been used to finance the government's treasury needs.

II. UNDERLYING DSA ASSUMPTIONS

5. **The current macroeconomic outlook takes into account changes in initial conditions (Text Table 2).** Inflation in 2010 was significant below its level previously projected (in the 2010 DSA) as a result of lower food and oil prices. Fiscal deficit was also below initial projections, due to an improved revenue performance and donor support despite a rise in total public expenditure in an elections year. Moreover, the external current account deficit narrowed less than expected due to the negative terms of trade shock driven by lower export prices for coffee, which accounts for about 70 percent of total exports.

Text Table 2. Burundi: Macroeconomic Assumptions and Outcomes, 2009-10

	2009		2010	
	2010 DSA	Act.	2010 DSA	Act.
	(Annual percentage change)			
National income and prices				
Real GDP growth	3.5	3.5	3.9	3.8
Consumer prices (end of period)	4.6	4.6	8.0	4.1
External sector				
	(Percent of GDP, unless otherwise indicated)			
General government				
Revenue (excluding grants)	13.6	13.6	13.5	14.6
Total expenditure and net lending	28.8	32.2	36.5	37.3
Overall balance				
Cash basis, after non-HIPC grants)	-3.8	-3.8	-3.7	-3.3
External sector				
Current account balance	-10.6	-11.5	-6.6	-9.9
Gross official reserves				
End of period (US$ million)	323.0	323.0	310.7	332.0
In months of imports of the following year	6.5	5.0	6.0	4.8

Sources: Burundian authorities and Fund staff estimates.

6. **Provided the security situation continues to improve, Burundi's medium and long-term outlook remains favorable (Box 1).** Real GDP growth is expected to average 6.7 percent over the long term, contingent upon continued consolidation of peace and security, a stable macroeconomic environment, and steadfast implementation of structural reforms in areas including public sector governance, business environment, as well as leveraging Burundi's integration to the East African Community (EAC) as a growth engine. Total investment is expected to average about 18 percent of GDP over the medium-term, a level considered supportive of economic growth. Fiscal policy will be supportive of economic growth and poverty reduction, while safeguarding debt sustainability. The fiscal deficit is expected to gradually decline during the later years. Revenue mobilization, reinforced by ongoing reforms being implemented by Burundi Revenue Authority, will lead to a gradual increase in government revenues, while expenditure will gradually decline to more sustainable levels. Exports of goods and services will be driven by investments in key export sectors over the medium term (coffee, tea, sugar, horticulture, and tourism). The external current account deficit, including grants, will decline over time, despite lower donor support. Financing will continue to be in the form of grants and highly concessional loans.

Investment spending will be critical for achieving higher growth. Important areas of investment include infrastructure renovation (power, transport, and communications) and export-generating activities such as tourism and mining. Key projects to be undertaken over the short to medium term include the development of Burundi's domestic hydroelectric potential (Kaganuzi, Mpanda, Kabu 16) and provision of adequate power lines with necessary substations to all provincial capitals. The rehabilitation and maintenance of national highways and community and rural road networks will also contribute to reducing Burundi's infrastructure gap. Implementing fundamental structural reforms—which has already begun—is needed to facilitate the necessary increase in private investment. These reforms include measures to improve the business regulatory environment and increase competition. There is scope for a significant increase in growth in the medium and long term, should potential investments in the mining sector materialize.[6]

7. **Three essential differences underpin the current macroeconomic framework compared to the one used in the previous LIC DSA: (**i) real GDP and export growth are higher over the medium term, reflecting the impact of reforms in the coffee sector, the spillovers from ongoing energy projects, and investments by the private sector on non-traditional sectors; (ii) official transfers are assumed to decline over the projection period, mostly due to policy changes in development partners assistance modalities; and (iii) nominal GDP is higher as a result of the rebasing of national accounts and also to the increase in inflation in 2011 as well.

8. **Risks to the macroeconomic outlook stem mostly from the security situation and the external environment.** The Euro Area debt crisis is likely to engender negative spillovers through the trade channel and is generating uncertainty in donor support. Supply shocks could lead to higher oil prices, further aggravating inflationary and budgetary pressures.

[6] As was mentioned in the 2010 DSA citing the World Bank's Country Economic Memorandum (2010), growth was expected to be stronger averaging 8.8 percent over 2011–15 if the various investments in infrastructure (including in roads and energy) in the amount of $556 million were materialized, and improved business environment and public finance management (PFM) reforms were accelerated and translated into a better FDI growth.

Box 1: Burundi: Rebasing of National Accounts

Rebased national accounts for 2005–08 were adopted by the government in November 2011. The revisions included improved data sources, a change in the base year from 1986 to 2005, and updated compilation methodologies (the 1993 UN System of National Accounts).

Under the new data, nominal national incomes have been revised up by an average of 36 percent during 2005–08, while the average real growth rate for these years is about 1.5 percentage point higher than earlier estimated. Using the rebased data, the gross national income per capita is estimated to have reached US$ 245 in 2010.

Official national accounts data have not been released since 1998 due to years of conflict and weak institutional capacity. Since 2006, Burundi has become a member of the *Observatoire économique et statistique d'Afrique Subsaharienne* (Afristat). This membership has benefitted Burundi in several regards, including in terms of technical assistance in updating national accounts. National account data for 2009 and 2010 is planned to be released in 2012.

Box 2: Burundi: Main Macroeconomic Assumptions, 2011–31

Real GDP growth is expected to accelerate to 6 percent over the medium term reflecting the consolidation of peace and security, the easing of infrastructure bottlenecks including higher electricity generation, accelerated implementation of other structural reforms including PFM, investment climate, coffee sector reform, and EAC integration. Average growth over the projection period is expected to be about 6.7 percent, compared with an average of 4.1 percent during 2008–10. Total investment is expected to average about 18 percent of GDP over the medium term, a level considered to be supportive of long-term economic growth. Key areas include infrastructure rehabilitation (power, transport, and communications) to address existing bottlenecks and export-generating activities such as tourism and mining.

Inflation is assumed to decline gradually over the medium term and stabilize at about 6 percent over the long term, in line with the EAC convergence criterion.

Fiscal policy would be supportive of economic growth and poverty reduction, while safeguarding macroeconomic stability in line with EAC convergence requirements and debt sustainability, which carries a high risk. Primary deficit is expected to average 2.3 percent of GDP over the medium term and then decline by 1 percentage point over the longer term.

- **Revenues**, excluding grants, are projected to stabilize at about 17 percent of GDP over the end of the projection period owing to the widening of the tax base through tax administration reforms.
- **Primary expenditure** is expected to gradually decline over time to about 30 percent of GDP by 2016. This path is consistent with a gradual reduction of current spending in favor of higher pro-poor spending. Public investment is expected to gradually decline over time to about 8 percent of GDP on average over the projection period while private investment gradually rises as the private sector becomes the engine of growth.

Exports of goods and services are projected to increase by an average of 7.6 percent over the medium term reflecting the impact of coffee sector reforms; they are projected to increase by 11.1 percent over the longer term as investment in new export sectors (horticulture, mining, and tourism) begins to yield results. In volume terms, exports of goods are projected to stabilize at about 7.5 percent over the medium term, before gradually declining to about 6 percent in the later years.

Imports of goods and services will accelerate from an average of about 5 percent over the medium term to about 8 percent over the longer term driven by strong growth in construction sector with the support of the Burundian Diaspora and as the tourism sector is developing. With emergency and reconstruction-related imports winding down, merchandise import volumes are expected to move broadly in line with real GDP growth in the later years.

The current account deficit, including grants, is expected to stay around 10 percent of GDP over the medium term. It will subsequently deteriorate as official transfers begin to wane, before declining gradually to about 7.5 percent at the end of the projection period. Excluding transfers, the current account deficit is expected to decline from 30 percent of GDP in 2011 to about 12 percent at the end of the projection period. The authorities will rely largely on grants and highly concessional loans to finance the current account deficit. This will require stepped up efforts by the authorities to implement governance reforms. The national governance strategy recently adopted by the Cabinet and the publication of voted and executed budgets demonstrate government's commitment to improved governance.

External financing will rely increasingly on loans in the long term, as grants' share of total financing gradually subsides. Grant support is assumed to decline from 24 percent of GDP in 2011 to about 13.4 percent in 2016 as humanitarian assistance gradually declines. Loans are assumed to gradually decline from about 3 percent of GDP in 2010 to about 1 percent in the later years. New borrowing (excluding EXIM Bank of India for the construction of Kabu 16 hydroelectric plant) is assumed exclusively on highly concessional terms with a grant element of 50 percent until 2017. From 2018 onward and as debt management capacity improve, one-third of the new financing will assumed on less concessional terms (35 percent grant element).

III. EXTERNAL DSA

9. **Under the baseline scenario, there is a breach of the PV of debt-to-exports indicative threshold.** The PV of debt-to-exports ratio will remain above the 100 percent threshold over the medium-term, mostly due to its narrow export base, and decline below its threshold toward the end of the projection period. In contrast, the PV of debt-to-GDP, the PV of debt-to-revenues, the debt service to exports ratio, and the debt service to revenue ratios are expected to remain well below the indicative policy dependent thresholds throughout the projection period (Text Table 3 and Table 1).

Text Table 3. Burundi: Summary of Baseline External Debt Sustainability Indicators

(percent)

	Indicative Threshold	2011	2021	2031	Average 2011-31
PV of debt to GDP	30	13.9	7.4	6.5	9.1
PV of debt to exports	100	174.4	108.1	88.9	123.2
PV of debt to revenue	200	86.8	52.4	53.3	62.5
Debt service to exports	15	3.5	7.9	3.9	6.9
Debt service to revenues	25	1.7	3.8	2.4	3.4

Source: Burundian authorities; Bank-Fund staff estimates and projections.

10. **Alternative scenarios and stress tests highlight the risk of further deterioration in the PV of debt-to-exports ratio.** The most severe breach of the indicative threshold for the PV of debt-to-exports ratio occurs under a decline in export value growth (Table 2, test B2). Under these circumstances, the PV of debt-to-exports ratio at the end of the projection period is higher by about 78 percentage points compared to the baseline scenario. This result underscores the importance of improvements in the business environment to encourage investment flows, and export diversification to reduce vulnerability to shocks. The alternative scenario based on new borrowing at higher interest rates (Table 2, scenario A2) shows that there is a protracted deterioration in Burundi's PV of debt-to-exports ratio compared to the baseline scenario. This result underscores the need for continued reliance on grants and highly concessional financing. Finally, the historical scenario (Table 2, scenario A1) suggests a sizable and protracted deterioration of the PV of debt-to-exports ratio. While important, the results of the historical scenario should be interpreted with caution, given Burundi's history of conflict.[7]

[7] This argument applies also to stress tests. In particular, the standardized stress test assuming export value growth at historical average minus one standard deviation in 2011–12 results in a

(continued)

11. **Alternative scenarios and stress tests suggest that the narrow export base is the most significant factor of vulnerability to Burundi's debt sustainability.** The indicative threshold for the debt service to exports ratio will be breached by a small margin in the medium-term under the combination stress test (Table 2, scenario B5). A similar breach occurs under the historical exports stress test (Table 2, scenario B2).[8] Finally, the indicative threshold for the PV of debt to GDP, PV of debt to revenue, and debt service to revenue ratios are not breached under any alternative scenario and stress test.

12. **In staffs' view Burundi faces a high risk of debt distress.** The preceding analysis suggests there is a breach of the PV of debt-to-exports threshold under the baseline scenario, while alternative scenarios and stress tests suggest that two debt burden thresholds are likely to be breached under less favorable scenarios and shocks.

13. **The authorities broadly share staffs' assessment.** They have expressed their commitment to continue to rely on highly concessional loans and grants to meet their financing needs, while advancing structural reforms to boost export capacity. They plan to request from the World Bank a Debt Management Performance Assessment (DeMPA) technical assistance that will prepare the groundwork for the development of a medium term debt strategy (MTDS).

IV. PUBLIC DSA

14. **Public debt indicators are expected to gradually improve under the baseline scenario.** The improvement is due primarily to a decline in the public sector borrowing requirement, reflecting the widening of the revenue base and the gradual decline in government spending in the post reconstruction period. The ratios of the PV of public debt to GDP and public debt to revenues remain low, reflecting Burundi's reliance on grants and highly concessional loans to finance reconstruction and poverty reduction.

15. **The alternative scenarios and stress tests show that public debt indicators can worsen rapidly, especially in the absence of fiscal consolidation, and lower economic growth.** Under the scenario of lower GDP growth (Table 4, scenario A3), the PV of-debt-to revenues ratio at the end of the projection period will worsen substantially compared to the baseline scenario. This result highlights the need for fiscal prudence and avoidance of past unsustainable borrowing policies. Preparation of a medium-term debt strategy will be essential

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sharp reduction of exports. The reason is the large standard deviation of exports reflecting the country's conflict, and export cyclicality (primarily owing to the old age of coffee trees).The year 2009 is excluded from the calculation of the 10-year standard deviation and historical average due to the accounting effects of the HIPC debt relief.

[8] See footnote 5.

for anchoring fiscal policy.[9] The debt service to revenue ratio is not significantly affected by alternative scenarios and shocks because additional borrowing is expected to be on highly concessional terms.

V. CONCLUSION

16. **Burundi will continue to face a high risk of debt distress**. The PV of external debt-to-exports ratio will exceed the country-specific debt burden threshold in the medium-term and is expected to be just below its threshold towards the end of the projection period. In addition, under alternative scenarios (including low growth and higher interest rate new borrowing) and stress tests two debt indicators exceed the country-specific debt burden thresholds. To mitigate risks to debt sustainability, staffs encourage the authorities to continue to rely on grants and highly concessional loans to meet financing needs, and further strengthen debt management. In addition, staffs advise the authorities to intensify structural reforms in order to diversify the export base, address existing infrastructure bottlenecks that hinder growth, and encourage private sector development by improving the business environment. Finally, staffs urge the authorities to continue efforts to improve tax administration and widen the tax base to reduce reliance on foreign financing.

[9] The caveat concerning the suitability of the historical scenario in a country marred by conflict applies.

Table 1.: External Debt Sustainability Framework, Baseline Scenario, 2008-2031 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections						2011-2016 Average			2017-2031 Average
	2008	2009	2010			2011	2012	2013	2014	2015	2016		2021	2031	
External debt (nominal) 1/	**101.6**	**20.7**	**22.5**			**22.6**	**22.5**	**22.7**	**22.6**	**20.4**	**18.3**		**12.1**	**10.6**	
o/w public and publicly guaranteed (PPG)	101.6	20.7	22.5			22.6	22.5	22.7	22.6	20.4	18.3		12.1	10.6	
Change in external debt	-17.6	-80.9	1.8			0.1	-0.1	0.2	-0.2	-2.1	-2.1		-0.8	0.1	
Identified net debt-creating flows	-29.8	-10.8	0.4			0.2	1.0	-2.5	-2.6	-3.6	-2.8		-1.6	-1.8	
Non-interest current account deficit	**1.2**	**11.0**	**9.8**	**6.3**	**6.3**	**12.4**	**11.9**	**9.1**	**9.0**	**10.7**	**11.2**		**10.8**	**7.5**	9.3
Deficit in balance of goods and services	-49.1	-47.9	-48.2			-45.4	-43.0	-38.9	-37.5	-35.7	-33.7		-31.0	-26.5	
Exports	9.8	6.5	8.9			8.0	8.7	8.7	8.6	7.7	6.9		6.8	7.3	
Imports	-39.3	-41.3	-39.2			-37.4	-34.3	-30.3	-29.0	-28.0	-26.8		-24.1	-19.2	
Net current transfers (negative = inflow)	-27.5	-23.8	-20.9	-18.6	5.5	-17.4	-14.1	-12.9	-11.7	-9.9	-9.0		-6.6	-4.7	-6.1
o/w official	-21.3	-15.0	-15.2			-12.2	-9.1	-8.1	-7.2	-5.7	-5.1		-3.8	-2.7	
Other current account flows (negative = net inflow)	77.9	82.6	78.9			75.3	69.0	60.9	58.2	56.3	53.8		48.3	38.7	
Net FDI (negative = inflow)	**-10.1**	**-7.8**	**-7.4**	**-5.1**	**3.5**	**-11.4**	**-10.0**	**-10.7**	**-10.4**	**-13.3**	**-13.0**		**-11.8**	**-8.5**	-10.5
Endogenous debt dynamics 2/	**-20.9**	**-13.9**	**-2.1**			**-0.8**	**-0.9**	**-0.9**	**-1.1**	**-1.1**	**-1.0**		**-0.6**	**-0.7**	
Contribution from nominal interest rate	0.7	0.5	0.1			0.0	0.1	0.1	0.1	0.1	0.1		0.1	0.1	
Contribution from real GDP growth	-4.9	-3.0	-0.7			-0.8	-1.0	-1.0	-1.2	-1.1	-1.1		-0.7	-0.8	
Contribution from price and exchange rate changes	-16.6	-11.5	-1.4			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	**12.2**	**-70.1**	**1.4**			**-0.1**	**-1.2**	**2.8**	**2.4**	**1.5**	**0.7**		**0.9**	**1.8**	
o/w exceptional financing	-0.3	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	13.5			13.9	14.1	14.5	14.6	13.1	11.7		7.4	6.5	
In percent of exports	...	...	150.8			174.4	162.1	167.2	170.0	171.0	169.4		108.1	88.9	
PV of PPG external debt	**...**	**...**	**13.5**			**13.9**	**14.1**	**14.5**	**14.6**	**13.1**	**11.7**		**7.4**	**6.5**	
In percent of exports	**...**	**...**	**150.8**			**174.4**	**162.1**	**167.2**	**170.0**	**171.0**	**169.4**		**108.1**	**88.9**	
In percent of government revenues	**...**	**...**	**92.3**			**86.8**	**85.3**	**86.6**	**88.4**	**82.0**	**75.0**		**52.4**	**53.3**	
Debt service-to-exports ratio (in percent)	**8.4**	**7.9**	**1.4**			**3.5**	**5.7**	**7.2**	**9.5**	**12.0**	**12.2**		**7.9**	**3.9**	
PPG debt service-to-exports ratio (in percent)	**8.4**	**7.9**	**1.4**			**3.5**	**5.7**	**7.2**	**9.5**	**12.0**	**12.2**		**7.9**	**3.9**	
PPG debt service-to-revenue ratio (in percent)	**5.9**	**3.8**	**0.9**			**1.7**	**3.0**	**3.7**	**4.9**	**5.8**	**5.4**		**3.8**	**2.4**	
Total gross financing need (Millions of U.S. dollars)	-125.4	66.8	52.7			28.8	61.0	-27.1	-20.6	-57.7	-40.3		-35.4	-139.3	
Non-interest current account deficit that stabilizes debt ratio	18.8	91.9	8.0			12.3	12.1	8.8	9.1	12.9	13.2		11.6	7.4	
Key macroeconomic assumptions															
Real GDP growth (in percent)	5.0	3.5	3.8	3.8	1.2	4.2	4.8	5.0	6.0	5.8	5.9	5.3	6.1	7.9	6.7
GDP deflator in US dollar terms (change in percent)	16.2	12.7	7.5	6.8	12.9	7.7	5.3	5.0	5.7	7.8	7.3	6.5	6.0	-1.0	3.7
Effective interest rate (percent) 5/	0.7	0.6	0.3	0.8	0.2	0.1	0.4	0.5	0.6	0.5	0.5	0.4	0.6	0.7	0.7
Growth of exports of G&S (US dollar terms, in percent)	55.4	-22.6	52.7	16.1	29.6	0.2	20.3	10.1	10.6	2.0	2.2	7.6	11.7	9.0	11.1
Growth of imports of G&S (US dollar terms, in percent)	31.7	22.7	5.8	21.7	27.2	7.0	1.2	-2.7	7.2	10.2	8.5	5.2	15.4	4.1	8.2
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	40.8	38.2	35.5	35.8	47.4	51.4	41.5	45.9	45.9	46.3
Government revenues (excluding grants, in percent of GDP)	13.9	13.6	14.6			16.0	16.5	16.7	16.5	16.0	15.6		14.1	12.2	13.5
Aid flows (in Millions of US dollars) 7/	38937.4	17681.6	37360.9			552.1	438.5	467.1	496.2	510.2	562.3		888.2	1815.4	
o/w Grants	438.0	1205.1	460.9			524.2	405.5	432.1	458.2	465.9	513.4		833.0	1699.2	
o/w Concessional loans	38499.3	16476.6	36900.0			27.9	33.0	35.0	38.0	44.2	48.9		55.2	116.2	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			24.0	17.2	16.7	15.8	13.9	13.4		11.7	9.8	11.1
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			94.9	91.5	89.6	89.9	94.7	95.8		95.1	95.0	95.1
Memorandum items:															
Nominal GDP (Millions of US dollars)	1556.1	1814.7	2024.0			2271.4	2506.7	2763.3	3095.2	3527.7	4007.7		7464.2	18199.8	
Nominal dollar GDP growth	22.0	16.6	11.5			12.2	10.4	10.2	12.0	14.0	13.6	12.1	12.5	6.8	10.6
PV of PPG external debt (in Millions of US dollars)			272.2			306.3	346.4	396.7	442.6	455.2	463.6		544.5	1164.5	
(PVt-PVt-1)/GDPt-1 (in percent)						1.7	1.8	2.0	1.7	0.4	0.2	1.3	0.4	0.5	0.4
Gross workers' remittances (Millions of US dollars)	…	…	…			…	…	…	…	…	…		…	…	
PV of PPG external debt (in percent of GDP + remittances)	...	...	13.5			13.9	14.1	14.5	14.6	13.1	11.7		7.4	6.5	
PV of PPG external debt (in percent of exports + remittances)	...	...	150.8			174.4	162.1	167.2	170.0	171.0	169.4		108.1	88.9	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	1.4			3.5	5.7	7.2	9.5	12.0	12.2		7.9	3.9	

Sources: Burundian authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2a.Burundi: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2011-2031
(In percent)

							Projections	
	2011	2012	2013	2014	2015	2016	**2021**	2031
PV of debt-to GDP ratio								
Baseline	14	14	15	15	13	12	**7**	6
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2011-2031 1/	14	13	15	17	18	18	**21**	18
A2. New public sector loans on less favorable terms in 2011-2031 2	14	14	16	16	15	14	**10**	11
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	14	14	15	15	13	12	**8**	7
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	14	15	18	17	16	14	**9**	7
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012-2013	14	15	18	18	16	14	**9**	8
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012-2013 4/	14	19	24	23	21	19	**12**	8
B5. Combination of B1-B4 using one-half standard deviation shocks	14	19	26	25	23	21	**13**	9
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	14	19	20	20	18	16	**10**	9
PV of debt-to-exports ratio								
Baseline	174	162	167	170	171	169	**108**	89
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2011-2031 1/	174	154	178	198	232	263	**300**	252
A2. New public sector loans on less favorable terms in 2011-2031 2	174	165	179	188	196	202	**150**	144
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	174	159	165	167	168	168	**106**	87
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	174	243	362	360	363	363	**230**	167
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012-2013	174	159	165	167	168	168	**106**	87
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012-2013 4/	174	219	278	272	275	276	**174**	107
B5. Combination of B1-B4 using one-half standard deviation shocks	174	245	342	335	338	339	**214**	136
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	174	159	165	167	168	168	**106**	87
PV of debt-to-revenue ratio								
Baseline	87	85	87	88	82	75	**52**	53
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2011-2031 1/	87	81	92	103	111	117	**145**	151
A2. New public sector loans on less favorable terms in 2011-2031 2	87	87	93	98	94	89	**73**	86
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	87	86	90	91	84	78	**54**	55
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	87	92	106	106	98	91	**63**	57
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012-2013	87	94	107	109	101	93	**65**	66
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012-2013 4/	87	115	144	142	132	122	**84**	64
B5. Combination of B1-B4 using one-half standard deviation shocks	87	116	156	153	143	132	**91**	71
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	87	117	120	121	113	104	**72**	73

Table 2b.Burundi: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2011-2031 (continued)
(In percent)

Debt service-to-exports ratio

Baseline	3	6	7	9	12	12	**8**	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2011-2031 1/	3	6	7	10	13	14	**11**	10
A2. New public sector loans on less favorable terms in 2011-2031 2	3	6	7	10	13	14	**8**	7
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	3	6	7	9	12	12	**8**	4
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	3	8	13	18	22	22	**15**	8
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012-2013	3	6	7	9	12	12	**8**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012-2013 4/	3	6	8	11	13	14	**10**	6
B5. Combination of B1-B4 using one-half standard deviation shocks	3	7	10	14	17	17	**13**	7
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	3	6	7	9	12	12	**8**	4

Debt service-to-revenue ratio

Baseline	2	3	4	5	6	5	**4**	2
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2011-2031 1/	2	3	4	5	6	6	**5**	6
A2. New public sector loans on less favorable terms in 2011-2031 2	2	3	4	5	6	6	**4**	4
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	2	3	4	5	6	6	**4**	2
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	2	3	4	5	6	6	**4**	3
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012-2013	2	3	5	6	7	7	**5**	3
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012-2013 4/	2	3	4	6	6	6	**5**	3
B5. Combination of B1-B4 using one-half standard deviation shocks	2	3	5	6	7	7	**6**	4
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	2	4	5	7	8	8	**5**	3
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	44	44	44	44	44	44	**44**	44

Sources: Burundian authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ The standarized stress test of assuming export value growth at historical average minus one standard deviation is not relevant for a country like Burundi which experienced conflict and whose exports suffer from cyclical factors.
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 3.Burundi: Public Sector Debt Sustainability Framework, Baseline Scenario, 2008-2031
(In percent of GDP, unless otherwise indicated)

	Actual					Estimate						Projections			
	2008	2009	2010	Average 5/	Standard Deviation 5/	2011	2012	2013	2014	2015	2016	2011-16 Average	2021	2031	2017-31 Average
Public sector debt 1/	114.8	35.3	37.8			37.0	35.9	35.6	34.7	32.3	29.9		22.7	18.8	
o/w foreign-currency denominated	101.6	20.7	22.5			22.6	22.5	22.7	22.6	20.4	18.3		12.1	10.6	
Change in public sector debt	-22.9	-79.5	2.4			-0.8	-1.1	-0.3	-0.8	-2.5	-2.3		-1.1	-0.1	
Identified debt-creating flows	-27.3	-111.0	-0.7			-2.0	-1.0	0.2	0.1	-1.7	-1.5		-0.5	0.0	
Primary deficit	-1.0	-44.1	2.3	-4.2	14.1	1.0	2.6	3.1	3.2	2.0	2.0	2.3	1.5	0.7	1.3
Revenue and grants	42.1	80.0	37.4			39.1	32.7	32.4	31.3	29.2	28.4		25.3	21.5	
of which: grants	28.2	66.4	22.8			23.1	16.2	15.6	14.8	13.2	12.8		11.2	9.3	
Primary (noninterest) expenditure	41.1	36.0	39.7			40.1	35.3	35.5	34.5	31.2	30.4		26.8	22.2	
Automatic debt dynamics	-24.0	-19.1	-3.0			-2.9	-3.4	-2.9	-3.0	-3.7	-3.4		-2.0	-0.7	
Contribution from interest rate/growth differential	-33.9	-15.3	-3.1			-4.5	-4.0	-3.0	-3.2	-3.6	-3.2		-2.1	-0.9	
of which: contribution from average real interest rate	-27.3	-11.4	-1.8			-2.9	-2.3	-1.3	-1.2	-1.7	-1.4		-0.7	0.5	
of which: contribution from real GDP growth	-6.6	-3.8	-1.3			-1.5	-1.7	-1.7	-2.0	-1.9	-1.8		-1.4	-1.4	
Contribution from real exchange rate depreciation	9.8	-3.8	0.1			1.5	0.6	0.1	0.2	-0.1	-0.2		...	...	
Other identified debt-creating flows	-2.2	-47.9	0.0			0.0	-0.2	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	-0.2	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	-2.2	-47.9	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	4.3	31.6	3.2			1.2	-0.1	-0.5	-1.0	-0.7	-0.8		-0.6	-0.1	
Other Sustainability Indicators															
PV of public sector debt	...	...	28.7			28.2	27.5	27.4	26.7	24.9	23.3		18.0	14.7	
o/w foreign-currency denominated	...	...	13.5			13.9	14.1	14.5	14.6	13.1	11.7		7.4	6.5	
o/w external	...	...	13.5			13.9	14.1	14.5	14.6	13.1	11.7		7.4	6.5	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	4.7	-39.9	5.7			6.4	8.1	9.9	10.8	10.2	10.8		11.4	9.1	
PV of public sector debt-to-revenue and grants ratio (in percent)	…	…	76.9			72.2	84.0	84.5	85.4	85.4	82.1		71.1	68.3	
PV of public sector debt-to-revenue ratio (in percent)	…	…	197.0			176.3	166.4	163.4	162.1	155.9	149.6		127.3	120.6	
o/w external 3/	…	…	92.3			86.8	85.3	86.6	88.4	82.0	75.0		52.4	53.3	
Debt service-to-revenue and grants ratio (in percent) 4/	4.1	1.5	1.8			4.3	5.7	6.7	6.9	7.3	6.8		5.1	4.1	
Debt service-to-revenue ratio (in percent) 4/	12.3	8.6	4.5			10.6	11.4	13.0	13.1	13.4	12.4		9.1	7.3	
Primary deficit that stabilizes the debt-to-GDP ratio	21.9	35.4	-0.1			1.8	3.7	3.4	4.0	4.5	4.3		2.6	0.8	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	5.0	3.5	3.8	3.8	1.2	4.2	4.8	5.0	6.0	5.8	5.9	5.3	6.1	7.9	6.7
Average nominal interest rate on forex debt (in percent)	0.7	0.6	0.3	0.8	0.2	0.1	0.4	0.5	0.6	0.5	0.5	0.4	0.6	0.7	0.7
Average real interest rate on domestic debt (in percent)	-16.5	-9.3	-3.2	-9.7	6.7	-3.4	-3.4	-0.7	-1.3	-3.2	-2.8	-2.5	-0.6	6.4	1.7
Real exchange rate depreciation (in percent, + indicates depreciation)	11.0	-4.3	0.3	4.9	15.4	7.9	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	27.3	16.9	7.5	12.6	13.0	9.6	10.8	7.6	8.5	10.6	9.4	9.4	8.2	1.2	5.9
Growth of real primary spending (deflated by GDP deflator, in percent)	0.1	-0.1	0.1	0.1	0.2	0.1	-0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.1	0.0
Grant element of new external borrowing (in percent)	...	...	...	…	…	40.8	38.2	35.5	35.8	47.4	51.4	41.5	45.9	45.9	46.3

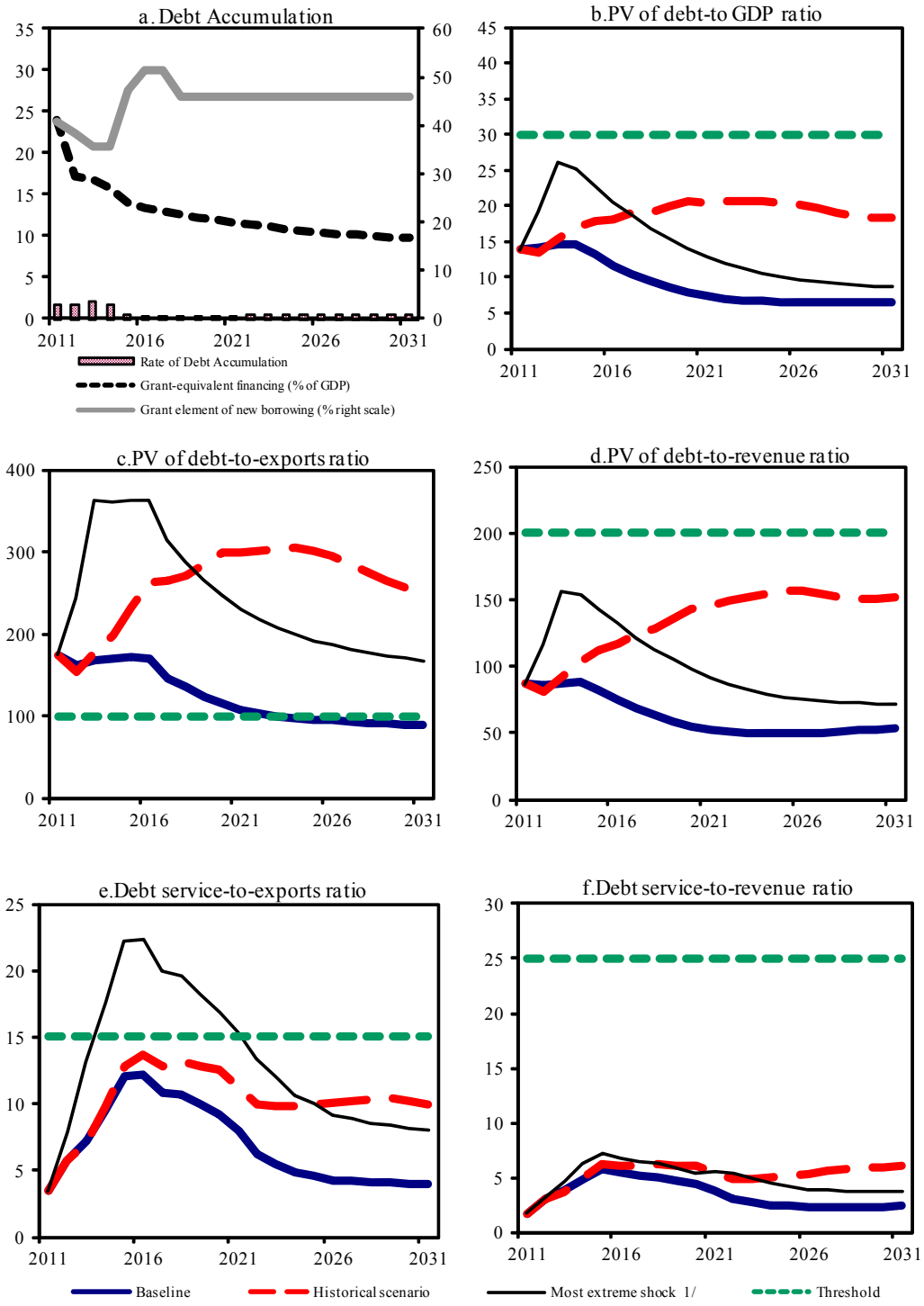
Sources: Burundian authorities; and staff estimates and projections.
1/ General government, on a gross basis.
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.
3/ Revenues excluding grants.
4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.
5/ Historical averages and standard deviations are derived over the past 10 years.

Table 4.Burundi: Sensitivity Analysis for Key Indicators of Public Debt 2011-2031

	Projections							
	2011	2012	2013	2014	2015	2016	2021	2031
PV of Debt-to-GDP Ratio								
Baseline	28	27	27	27	25	23	18	15
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	28	26	25	24	21	20	13	12
A2. Primary balance is unchanged from 2011	28	27	25	24	22	20	14	13
A3. Permanently lower GDP growth 1/	28	28	28	27	25	24	19	18
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2012-2013	28	28	29	29	27	26	21	19
B2. Primary balance is at historical average minus one standard deviations in 2012-2013	28	27	26	26	24	23	17	14
B3. Combination of B1-B2 using one half standard deviation shocks	28	27	26	26	25	23	19	18
B4. One-time 30 percent real depreciation in 2012	28	33	32	30	28	26	19	15
B5. 10 percent of GDP increase in other debt-creating flows in 2012	28	33	32	31	29	27	20	16
PV of Debt-to-Revenue Ratio 2/								
Baseline	72	84	84	85	85	82	71	68
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	72	80	77	74	71	66	49	43
A2. Primary balance is unchanged from 2011	72	81	78	76	74	70	56	58
A3. Permanently lower GDP growth 1/	72	84	85	86	87	84	75	83
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2012-2013	72	86	88	91	92	89	83	88
B2. Primary balance is at historical average minus one standard deviations in 2012-2013	72	83	81	82	82	79	69	67
B3. Combination of B1-B2 using one half standard deviation shocks	72	82	80	82	83	81	75	81
B4. One-time 30 percent real depreciation in 2012	72	101	98	97	96	91	74	68
B5. 10 percent of GDP increase in other debt-creating flows in 2012	72	101	100	100	100	96	81	74
Debt Service-to-Revenue Ratio 2/								
Baseline	4	6	7	7	7	7	5	4
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	4	6	7	7	7	7	5	4
A2. Primary balance is unchanged from 2011	4	6	7	7	7	7	5	4
A3. Permanently lower GDP growth 1/	4	6	7	7	7	7	5	5
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2012-2013	4	6	7	7	8	7	5	5
B2. Primary balance is at historical average minus one standard deviations in 2012-2013	4	6	7	7	7	7	5	4
B3. Combination of B1-B2 using one half standard deviation shocks	4	6	7	7	7	7	5	4
B4. One-time 30 percent real depreciation in 2012	4	6	8	8	9	8	6	5
B5. 10 percent of GDP increase in other debt-creating flows in 2012	4	6	7	7	8	7	5	4

Sources: Burundian authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

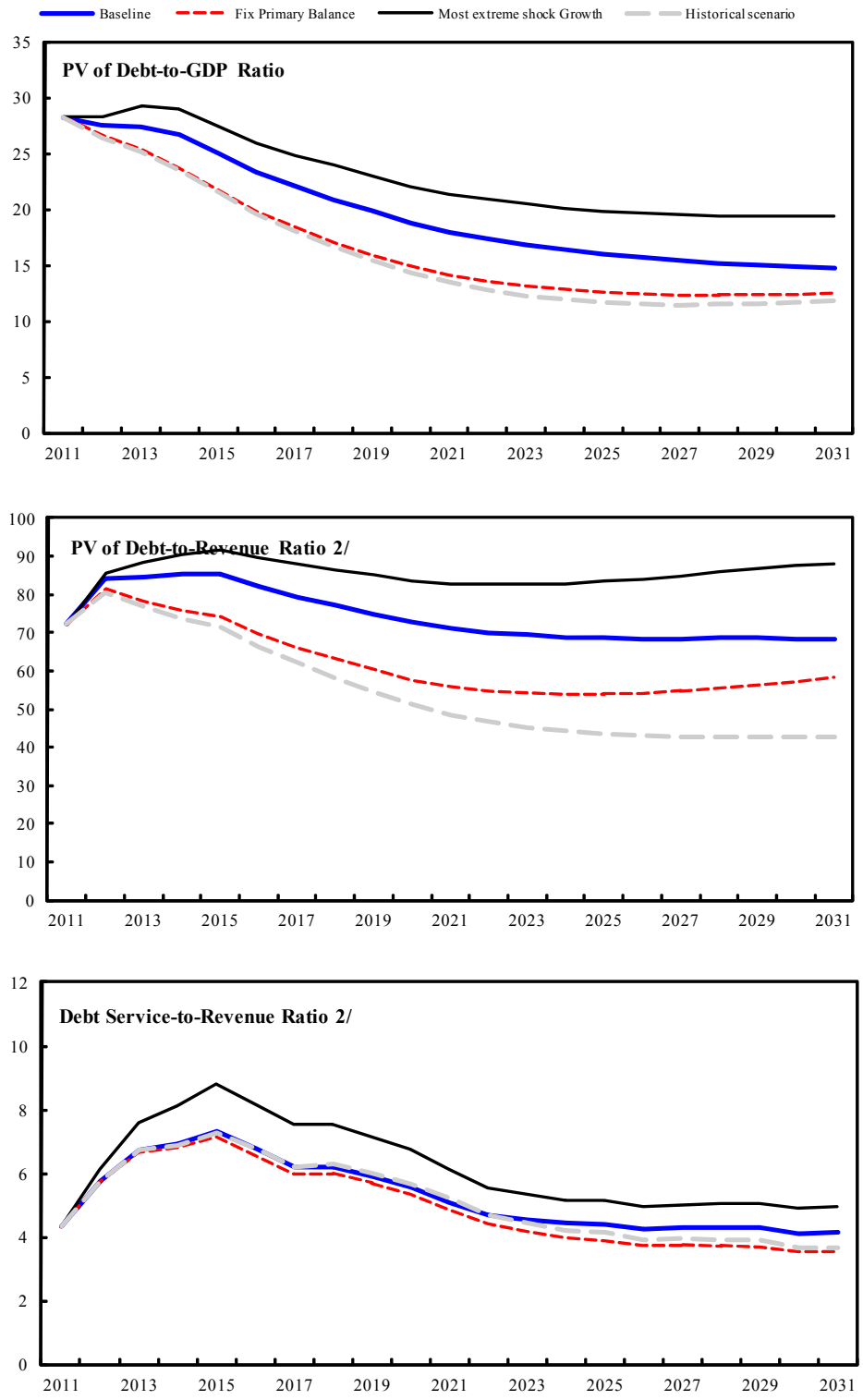
Figure 1. Burundi: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2011-2031 1/



Sources: Burundian authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2021. In figure b. it corresponds to a Combination shock; in c. to a Exports shock; in d. to a Combination shock; in e. to a Exports shock and in figure f. to a Combination shock

Figure 2.Burundi: Indicators of Public Debt Under Alternative Scenarios, 2011-2031 1/



Sources: Burundian authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2021.
2/ Revenues are defined inclusive of grants.